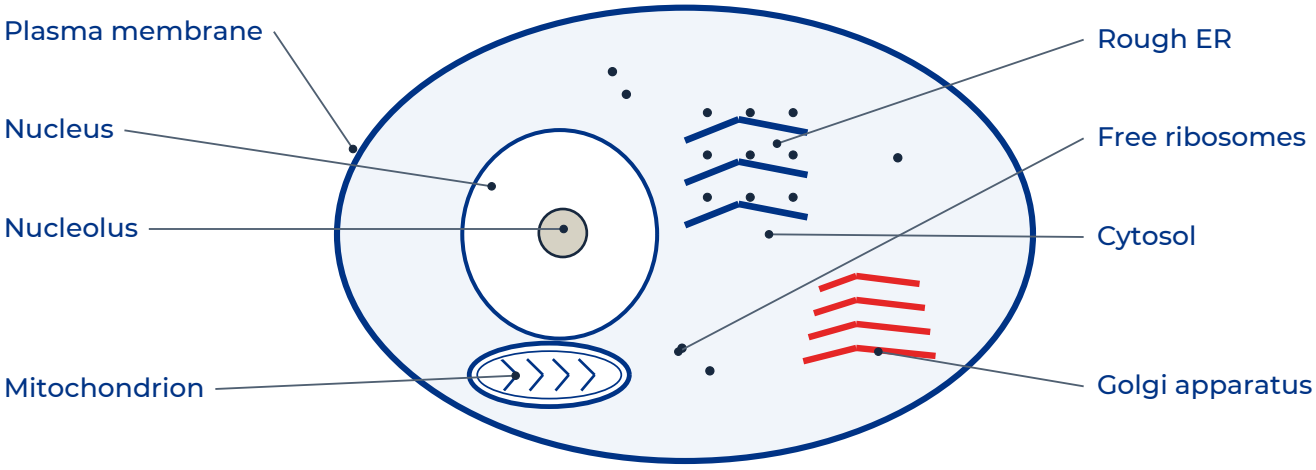
Animal cell structures

A generalized animal cell: identify the parts and connect structure to function.



Structure	Main role / useful distinction
Nucleus / nucleolus	Nucleus contains most cellular DNA. Nucleolus makes/processes rRNA and assembles ribosomal subunits.
Ribosomes / rough ER	Ribosomes translate mRNA. ER-associated ribosomes make ER-targeted proteins; rough ER helps process them.
Golgi / vesicles	Golgi modifies and sorts cargo. Vesicles carry material between compartments or to the cell surface.
Mitochondria	Inner-membrane cristae support oxidative phosphorylation. Most eukaryotic respiration stages occur here.
Cytosol / cytoplasm	Cytosol is the fluid. Cytoplasm includes that fluid and structures outside the nucleus.

Model limit: organelle sizes, counts and colors are schematic. Cell types and states differ.

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Self-check questions and explanations. Refer to the visual guide on the front.

What identifies the major cell structures in the diagram?

Membrane: outer boundary. Nucleus: DNA compartment. Nucleolus: dense nuclear region. Mitochondrion: folded inner membrane. Rough ER: membrane + ribosomes. Golgi: stacked sacs. Free ribosomes: cytosolic particles. Cytosol: intervening fluid.

Which structures have a membrane, and which do not?

For example, nucleus versus nucleolus: the nucleus is enclosed by a double nuclear envelope; the nucleolus is a non-membrane-bound region within it. Ribosomes are also non-membranous.

How do cytosol and cytoplasm differ?

Cytosol is the fluid portion outside organelles. In a eukaryotic cell, cytoplasm includes cytosol plus the structures outside the nucleus.

Which ribosomes make a secreted protein versus a cytosolic protein?

Ribosomes assemble polypeptides. A ribosome synthesizing an ER-targeted secreted protein associates with rough ER; free and bound ribosomes are not fundamentally different machines.

How do nucleus, mitochondrion and Golgi structure support their jobs?

Nuclear envelope: controls nuclear-cytosolic exchange through pores. Cristae: enlarge inner-membrane area for oxidative phosphorylation. Golgi sacs: enable processing and sorting through compartments.

Does every animal cell contain exactly one of each structure?

Numbers and shapes vary with cell type and state, and some specialized cells lack particular structures. The drawing cannot show molecular scale, dynamics or real organelle abundance.